

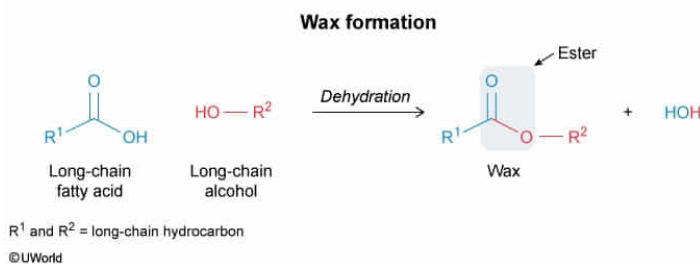
A wax is made up of a long-chain fatty acid and a long-chain alcohol joined by:

- ☐ A. an amide linkage. [1%]
- ✓ ☒ B. an ester linkage. [88%]
- ☐ C. an anhydride linkage. [3%]
- ☐ D. a glycosidic linkage. [6%]

Correct

88% answered correctly

Explanation:



Lipids are hydrophobic molecules that are classified based on characteristics such as length, linkage type, and solubility. These molecules are broadly classified as **hydrolyzable** (lipid contains hydrolyzable ester or amide linkages) or **nonhydrolyzable** (lipid does not contain hydrolyzable ester or amide linkages). The hydrolyzable lipids, such as phospholipids, glycolipids, and waxes, are further classified based on the structure of their backbone (glycerophospholipids, glycosphingolipids, etc.).

Waxes are an example of a hydrolyzable lipid that serves as a form of protection in plants and animals. Waxes are made up of a **long-chain fatty acid** and a **long-chain alcohol**. The wax forms when the acid and the alcohol combine through dehydration to form an **ester bond**.

(Choice A) Amide linkages are present in sphingolipids but not in waxes. For an **amide to form**, the carboxyl group of the fatty acid would need to combine with an amine rather than an alcohol.

(Choice C) Like esters, anhydrides are carboxylic acid derivatives. However, for an anhydride to form, the carboxyl group from the fatty acid would need to combine with an acid chloride rather than an alcohol.

(Choice D) A **glycosidic bond** joins a sugar to another molecule. Because waxes are composed of a fatty acid and an alcohol and do not contain sugars, a glycosidic bond cannot be present.

Educational objective:

Lipids are hydrophobic molecules broadly classified as hydrolyzable or nonhydrolyzable, and more specifically classified based on their backbone structure. Waxes are hydrolyzable lipids that contain an ester bond formed by the linkage between a long-chain fatty acid and a long-chain alcohol.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402043

System:
Functional Groups and Their Reactions